Missing values and results — answers

Answers to the 16 items in `missing value and result.pdf`, prepared 2026-10-05, updated 2026-11-07.

> **Read this first.** This laptop is not the machine that trained the models. The committed logs show training ran on another PC (user `MustafaJK`, RTX 3050). This laptop (RTX 5060) has no `progress.json`, no checkpoints and no regressor weights. 5 items are fully answered here (7, 8, 9, 15, 16), 2 are answered by a faithful reproduction on this laptop (5, 6), 5 are partly answered, and the rest need one script run on the training PC.

Status at a glance

#	Item	Status
1	Per-epoch training history	Partly — B, C, F, G rebuilt from logs; A missing
2	Model A epoch timing	Estimate only
3	Model A generator-update count	Needs training PC
4	Batch size for F and G	Very likely 32, not confirmed
5	Landmark regressor validation error	Done by reproduction on this laptop
6	Regressor training time	Reproduction (CPU) plus bound for the original
7	Native pixel size	Done
8	Per-class image counts	Done
9	Qualitative visual description	Done
10–12	CPU, RAM, CUDA, cuDNN, driver, WSL, Windows	This laptop only; training PC unverified
13	Frozen dependency lists	This laptop only
14	MediaPipe version, dataset source and date	Source known; version and date unconfirmed for training PC
15	Fixed-seed evaluation pass	Done (this laptop, seed 42)
16	Full per-class accuracy, all five models	Done

Files and numbers from the run

1. Per-epoch training history partly done

All 50 epochs for B, C, F and G were rebuilt from the committed training logs into `reports/paper/results/history_from_logs/loss_history_{B,C,F,G}.csv`. New charts `reports/paper/charts/loss_model_F.png` and `loss_model_G.png` show the adversarial and landmark terms on separate panels. Model A's history is not recoverable here: its log was already overwritten in every commit.

2. Model A epoch timing estimate

`driver.log` shows the last real A run from 12:30:13 to 19:46:59 on 2026-07-06, which is 26,206 s. Subtracting about 200 s of data loading gives roughly **26,000 s, about 7.2 h, about 520 s per epoch**. This assumes that launch ran all 50 epochs from scratch. The minimum–maximum range is not recoverable.

Measured totals for the other four models, from their logs:

Model	Mean s/epoch	Range (s)	Total s	Total h
B	538.8	417.4–574.4	26,941.4	7.48
C	232.4	224.5–261.9	11,618.5	3.23
F	239.2	235.7–261.9	11,962.5	3.32
G	289.5	283.2–308.4	14,475.1	4.02

G also spent 818 s training its auxiliary classifier before the GAN epochs.

3. Model A generator-update count needs training PC

Not recoverable here. For reference, Model B's log shows 2 generator updates per step in epochs 1–11 and 3 in epochs 12–50. A may differ.

4. Batch size for F and G likely 32

The method suggested in the PDF will not work: `progress.json` for F and G does not record steps per epoch. Indirect evidence points to 32: Model C was launched by a committed script with no override (so 32), and F's epoch time is within 5% of C's on the same network (236 s against 226 s). At 32 that is 1,520 steps per epoch on 48,645 training images. Shell history on the training PC is the only direct proof.

5. Landmark regressor validation error done by reproduction

The original weights are not on this laptop, so the regressor was **reproduced** here on 2026-11-07 with the repository's own code: landmarks re-extracted with MediaPipe 0.10.14, then the regressor trained with `train_regressor()` from `src/train_model_b.py` (seed 42, same 20-epoch schedule and callbacks).

The reproduction matches the original pipeline closely: hand detection is **27.45%** (14,834 of 54,049 images), exactly the rate the training PC recorded. Validation set: the last 10% of the detected images (1,484 images; 13,350 for training), as Keras `validation_split` does in the original.

Validation error	Regressor	Constant baseline (training mean)	Reduction
MSE, normalised (all 63 values)	0.00295	0.00767	61%
MAE, normalised (all 63 values)	0.0318	0.0561	43%
MSE, pixels² (x, y only)	72.6	188.5	61%
MAE, pixels (x, y only)	6.03	10.74	44%
Mean keypoint distance, pixels	9.49	16.87	44%

- **Reading it:** on a 128 × 128 image, a predicted keypoint lands on average 9.5 px from the MediaPipe keypoint, against 16.9 px for always predicting the average hand. The regressor clearly learned hand-specific structure, but its error is far from zero, which limits how precise the landmark supervision for B, F and G can be.
- **Pixel units:** pixel errors use only x and y (scaled by 128); MediaPipe's z has no pixel meaning.
- **Consistency check:** the original B training log shows a landmark loss of about 0.003 on generated images, the same scale as this validation MSE.
- **Caveat for the manuscript:** these numbers come from a retrained regressor, not the exact weights used for F and G. Running `collect_training_machine_facts.sh` on the training PC gives the original weights' numbers.

File: `reports/paper/results/landmark_regressor_eval_repro.json`; log: `reports/paper/logs/repro_landmark_regressor.log`.

6. Regressor training time reproduction plus bound

- **Reproduction on this laptop (CPU):** 560.9 s (9.3 min) for 20 epochs. The learning rate was halved at epochs 9 and 18, and the best epoch was 19.
- **Original on the training PC (GPU):** not recorded. Model B's final launch loaded the regressor already trained, so it was trained in one of two earlier short launches (319 s and 526 s, each including about 3 minutes of data loading). That puts it **under roughly 6 minutes**.
- **Landmark extraction:** 3,000.6 s on the training PC; on this laptop the run was interrupted once; the resumed part took 2,442 s for the last 39,049 images, which scales to about 3,400 s for all 54,049.

7. Native pixel size done

The images are not all the same size. All are 8-bit grayscale PNG.

Size	Images	Where
64 × 64	53,401	all classes
256 × 256	638	kaaf (320), meem (318)
768 × 1024	10	haa

8. Per-class image counts done

Total 54,049 images in 32 classes. Largest class is `ain` (2,114), smallest is `yaa` (1,293): an imbalance ratio of 1.63. Mean 1,689, standard deviation 184, coefficient of variation 0.109.

Class	n	Class	n	Class	n	Class	n
ain	2114	gaaf	1705	laam	1832	taa	1838
al	1343	ghain	1977	meem	1765	thaa	1766
aleff	1672	ha	1592	nun	1819	thal	1582
bb	1791	haa	1526	ra	1659	toot	1791
dal	1634	jeem	1552	saad	1895	waw	1371
dha	1723	kaaf	1774	seen	1638	ya	1722
dhad	1670	khaa	1607	sheen	1507	yaa	1293
fa	1955	la	1746	ta	1816	zay	1374

9. Qualitative description of the visual differences done

From viewing `real_vs_generated.png` and the five `samples_*.png` grids:

- **Models A and B (noise only).** Outputs are hand-like, with recognisable pose for open-hand letters such as *al*, *bb*, *la* and *sheen*. Fingers are soft and often merged or missing. Closed-fist letters (*fa*, *gaaf*, *ha*, *khaa*) collapse into fragmentary dark masses. Backgrounds are uniformly light. Pose varies widely between samples of the same letter. B shows no visible improvement over A.
- **Model C.** Outputs follow the reference hand's silhouette with sharp contours. Periodic checkerboard artefacts appear in the background: scattered patches on light backgrounds, and across the whole frame on dark or textured ones (*dha*, *ghain*, *ha*, *khaa*, *laam*, *thal*).
- **Model F.** Same structure as C. The artefacts are largely gone on light backgrounds but remain on dark or textured ones.

- **Model G.** The cleanest of the five. Only faint residual speckle remains on a few letters (*ha*, *khaa*, *laam*, *thal*, *aleff*).
- **C, F and G together.** The eight samples per letter are visually near-identical. The noise vector has little visible effect, so variety comes from the structure map, not from the noise.

Environment and software

10–12. Hardware and versions this laptop only

These values describe this laptop, not the PC that trained the models.

	This laptop
CPU / RAM	Intel Core i7-14650HX, 15.7 GB
GPU / driver	RTX 5060 Laptop 8 GB, driver 610.62
Windows	11 Pro, build 26200
WSL	2.7.10.0, Ubuntu 26.04 LTS, kernel 6.18.33.2
Python / TensorFlow / Keras	3.11.15 / 2.21.0 / 3.15.0
CUDA / cuDNN wheels loaded	CUDA 12.9.x, cuDNN 9.24.0.43

Two clues about the provisional line in the manuscript:

- "CUDA 12.5.1, cuDNN 9" is exactly what TensorFlow 2.21 reports as its build-time versions. That line may be the build string, not the libraries actually loaded.
- A comment in `src/train_model_b.py` says "avoids OOM at 16 GB", which conflicts with "32 GB RAM".

13. Frozen dependency lists this laptop only

Saved as `reports/paper/results/seed42/requirements_tf_eval_machine.txt` and `requirements_mp_eval_machine.txt`. Protobuf is 7.35.1 in the TensorFlow environment and 4.25.9 in the MediaPipe one, matching the "4.x vs 7.x" note in the code. The training PC's lists still need to be captured.

14. MediaPipe version and dataset source partly done

MediaPipe is 0.10.14 on this laptop; the training PC's version is unconfirmed. The dataset is `pain/ArASL_Database_Grayscale` on Hugging Face. The local parquet file is dated 2026-06-29; the download date on the training PC is unknown.

Additional runs

15. Fixed-seed evaluation pass done

Run on this laptop (CPU) on 2026-11-07 with seed 42. One reference classifier was trained, saved and used by *both* `paper_eval.py` and `confusion_matrix.py`. Both logs print the same provenance line:

```
path=reports/paper/results/seed42/reference_classifier_seed42.keras
mtime=2026-11-07 18:06:09   seed=42   samples_per_class=40 (paper_eval) / 60 (confusion_matrix)
reference classifier on real held-out = 0.9754
```

Model	Recognition	Diversity	SSIM	Held-out recognition	Generalisation gap	Confusion-matrix accuracy
A	0.6211	0.1779	—	—	—	0.6323
B	0.6484	0.1926	—	—	—	0.6594
C	0.7172	0.4134	0.7521	0.6969	0.0203	0.7224
F	0.8609	0.3722	0.8249	0.8370	0.0240	0.8656
G	0.9383	0.4010	0.8251	0.9062	0.0321	0.9380

- **Why two recognition columns:** both come from the same classifier, but `paper_eval.py` scores 40 samples per class and `confusion_matrix.py` scores 60, with different noise seeds. They differ by less than 0.012.
- **New numbers replace the old:** they are slightly lower than the earlier unseeded logs (for example G 0.9383 against 0.9461), because the classifier is a different instance. The ranking A < B < C < F < G is unchanged. The manuscript's headline table and Table 17 should both take their values from this run.
- **Consistency check:** diversity for A, B, F and G matches the original run to four decimals, so the generators behave identically on this laptop.
- **Reproducibility note:** training was interrupted by low memory and resumed from saved epoch 11 (re-seeded with 42 + 11). Re-running gives the same result, but it is not bit-identical to an uninterrupted run.

Files: `reports/paper/results/seed42/metrics.json`, `metrics.csv`, `confusion_[A-G].png`, `confusion_summary.json`; logs in `reports/paper/logs/*_seed42.log`.

16. Full per-class accuracy, all five models done

Fraction of the 60 generated images per letter that the reference classifier assigned to the intended letter (from `reports/paper/results/seed42/per_class_accuracy.csv`).

Class	A	B	C	F	G

ain	0.833	0.800	0.667	0.850	0.833
al	0.967	0.967	0.967	0.967	0.983
aleff	0.417	0.367	0.333	0.650	0.833
bb	0.817	0.783	0.517	0.817	0.883
dal	0.633	0.383	0.883	0.883	0.933
dha	0.717	0.583	0.667	0.833	0.933
dhad	0.817	0.850	0.767	0.833	0.933
fa	0.450	0.683	0.483	0.783	0.883
gaaf	0.100	0.300	0.533	0.833	0.933
ghain	0.333	0.467	0.700	0.883	0.983
ha	0.733	0.383	0.400	0.567	0.933
haa	0.700	0.567	0.767	0.883	0.933
jeem	0.250	0.500	0.783	0.900	0.967
kaaf	0.483	0.617	0.617	0.783	0.917
khaa	0.500	0.617	0.850	0.950	1.000
la	0.767	0.883	0.833	0.867	0.967
laam	0.767	0.750	0.817	0.833	0.883
meem	0.817	0.800	0.650	0.850	0.950
nun	0.817	0.850	0.867	0.983	0.983
ra	0.617	0.633	0.633	0.833	0.917
saad	0.650	0.717	0.633	0.883	0.967
seen	0.850	0.800	0.883	0.983	1.000
sheen	0.900	0.933	0.983	1.000	1.000
ta	0.800	0.867	0.900	0.950	1.000
taa	0.700	0.600	0.633	0.833	0.917
thaa	0.167	0.433	0.583	0.833	0.933
thal	0.450	0.517	0.833	0.933	1.000
toot	0.600	0.733	0.750	0.917	0.967

waw	0.533	0.633	0.767	0.917	0.917
ya	0.867	0.700	0.783	0.933	0.900
yaa	0.950	0.817	0.983	1.000	1.000
zay	0.233	0.567	0.650	0.733	0.833
mean	**0.632**	**0.659**	**0.722**	**0.866**	**0.938**

Three largest confusions per model (intended → predicted, share of that letter's samples):

Model	Top confusions
A	gaaf → fa (0.483); thaa → taa (0.433); ghain → ain (0.383)
B	gaaf → ha (0.300); ha → fa (0.267); haa → khaa (0.250)
C	kaaf → seen (0.200); aleff → khaa (0.183); saad → ya (0.183)
F	kaaf → seen (0.217); ha → kaaf (0.100); ra → toot (0.100)
G	aleff → nun (0.067); kaaf → seen (0.067); aleff → ghain (0.050)

What to run on the training PC

```
bash collect_training_machine_facts.sh
```

This copies the five histories, summarises Model A's timing and update counts, evaluates the original regressor against a constant baseline, records the environment and writes both `pip freeze` files. It has not been run, because the files it reads are not on this laptop.

Files added or changed

- `src/paper_eval.py`, `src/confusion_matrix.py` — seeded, shared classifier, full per-class output
- `src/eval_landmark_regressor.py` — regressor validation error and baseline
- `src/paper_charts_from_logs.py` — F and G loss charts
- `collect_training_machine_facts.sh` — one-shot collector for the training PC
- `reports/paper/results/history_from_logs/` — four loss histories and `timing_summary.json`
- `reports/paper/charts/loss_model_F.png`, `loss_model_G.png`
- `reports/paper/results/seed42/` — seeded evaluation results, per-class accuracy, confusion matrices, saved classifier, this laptop's dependency lists

- `src/data.py`, `src/train_model_c.py` — image loader writes into one preallocated array (same values, about half the peak memory)
- `run_missing_values.sh` — resumable run for items 5, 6, 15 and 16
- `experiments/scripts/extract_lm_full.py` — reads images memory-mapped (same values, far less memory)
- `reports/paper/results/landmark_regressor_eval_repro.json` — reproduced regressor error and baseline